

STACK

Q3)

Question:

DS	Session	Stack	Question Information	Level 1	Challenge 43
Problem Description: You are given an array A of Q integers and Q queries. In each query, you are given an integer i ($1 \leq i \leq N$). Your task is to find the minimum index greater than i ($1 \leq i \leq N$) such that: 1. Sum of digits of A_i is greater than the sum of digits of A_j 2. $A_i < A_j$ If there is no answer, then print -1. Constraints $1 \leq N, Q \leq 10^5$ $1 \leq A_i \leq 10^9$ $1 \leq Q_i \leq N$ Input format • The first line contains two numbers N and Q. • The next line contains N numbers. • Next Q lines contain Q queries. Output format Print the answer as described in the problem					

PROGRAM:

```
1  #include <stdio.h>
2
3  int sumDigits(int n) {
4      int sum = 0;
5
6      while (n > 0) {
7          sum += n % 10;
8          n /= 10;
9      }
10
11     return sum;
12 }
13
14 int main() {
15     int n, q, i, j, x;
16     int a[100005];
17     int ans[100005];
18
19     scanf("%d %d", &n, &q);
20
21     for (i = 0; i < n; i++)
22         scanf("%d", &a[i]);
23
24     for (i = 0; i < q; i++) {
25         scanf("%d", &x);
26         x--;
27
28         ans[i] = -1;
29
30         for (j = x + 1; j < n; j++) {
31             if (a[x] < a[j] && sumDigits(a[x]) > sumDigits(a[j])) {
32                 ans[i] = j + 1;
33                 break;
34             }
35         }
36     }
37
38     printf("\n");
39
40     for (i = 0; i < q; i++) {
41         if (i > 0)
42             printf(" ");
43         printf("%d", ans[i]);
44     }
45
46     return 0;
47 }
```

OUTPUT:

TEST CASE 1:

```
5 5
62 70 28 62 92
1
5
3
4
2
2 -1 4 -1 -1
```

TEST CASE 2:

```
6 6
12 71 21 16 91 11
11
15
31
14
21
6
-1 -1 -1 -1 -1 -1
```